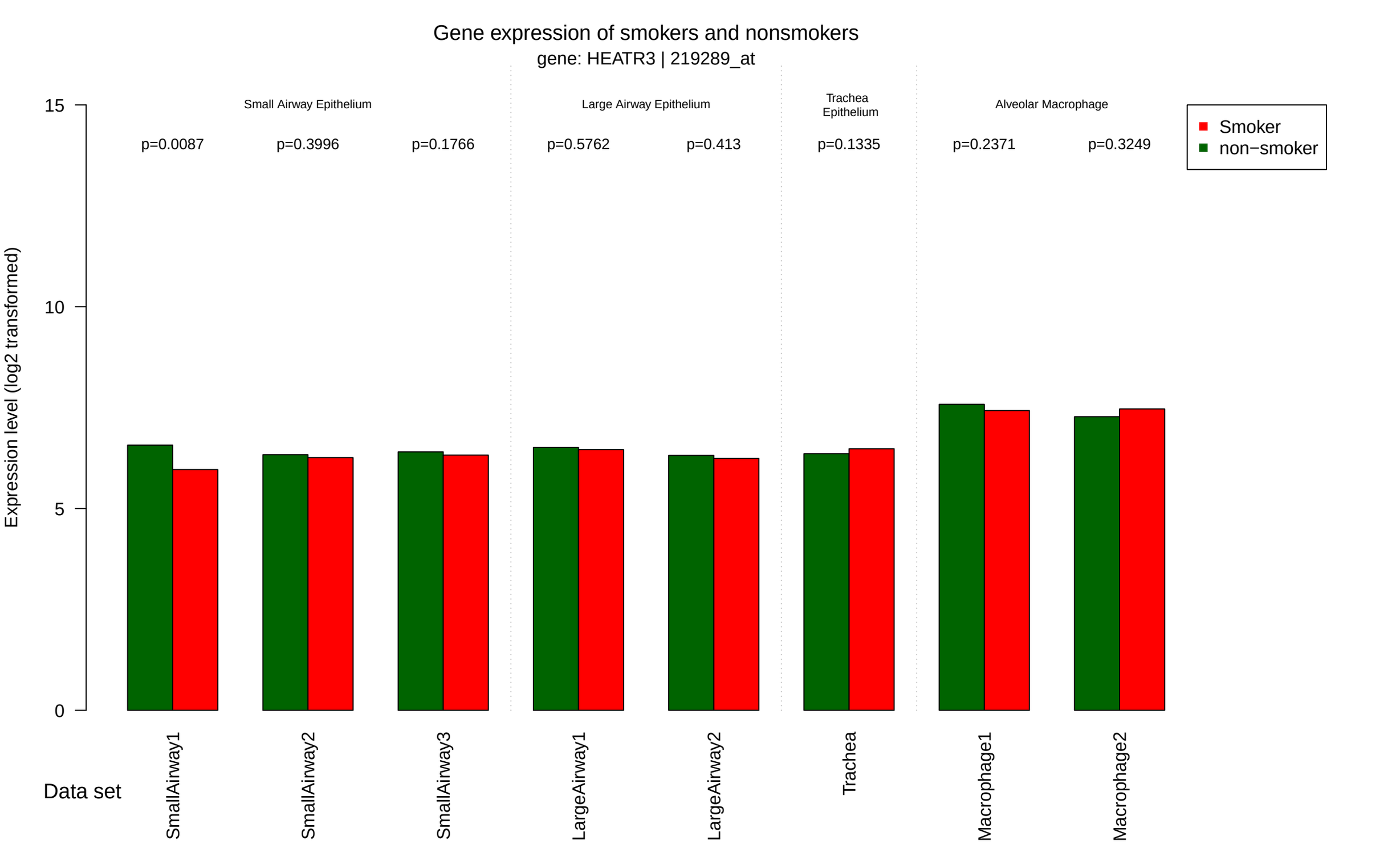
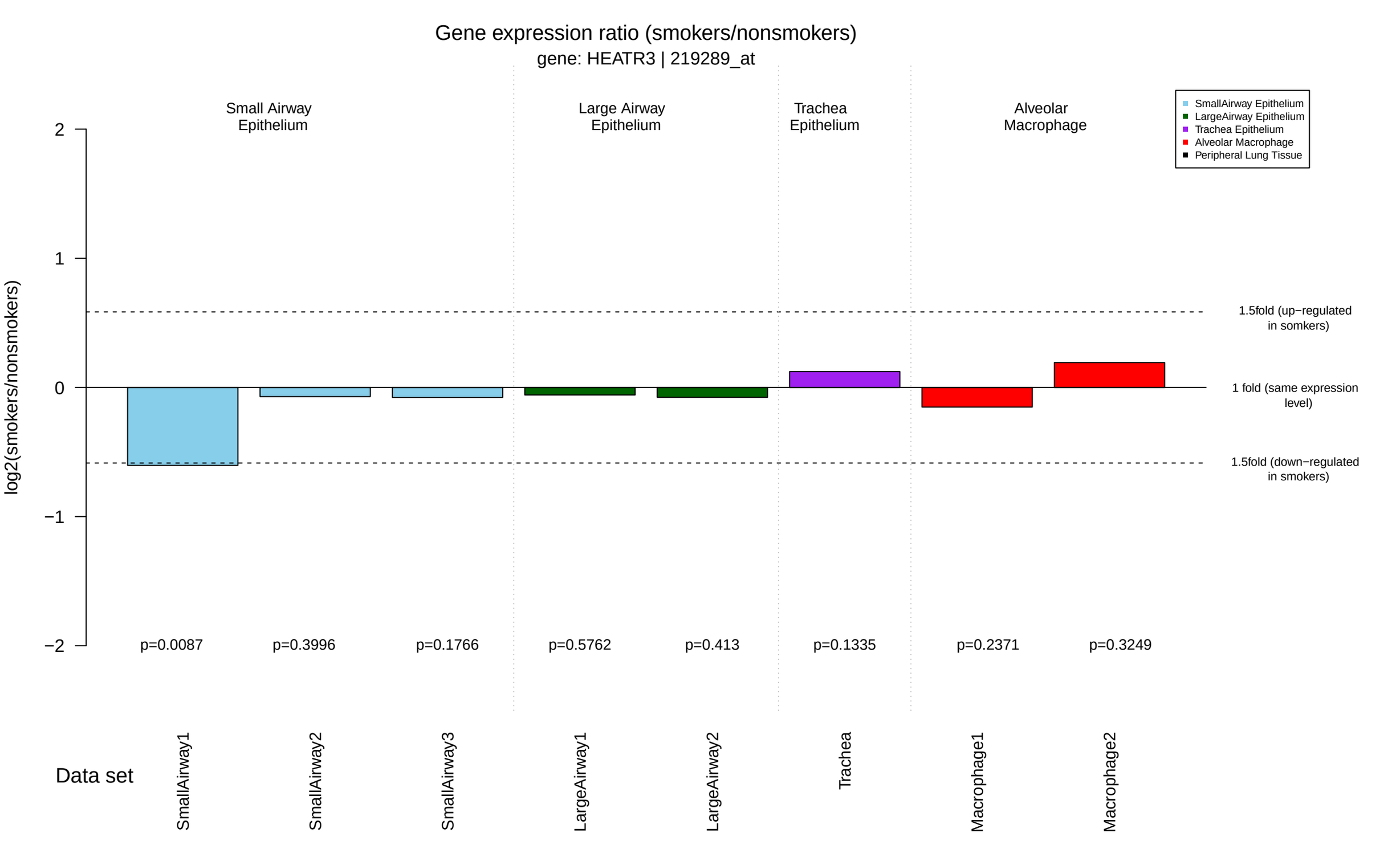
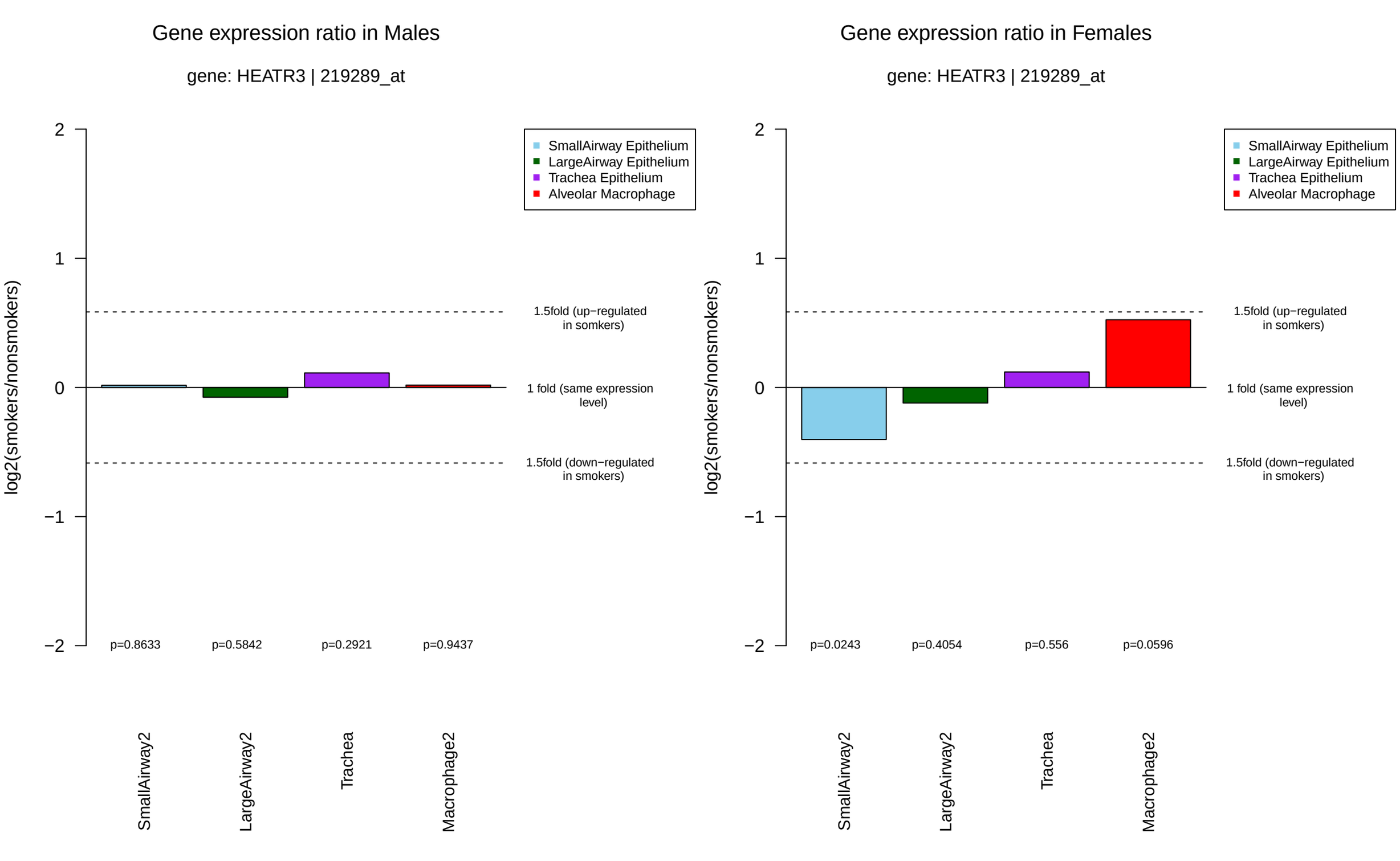
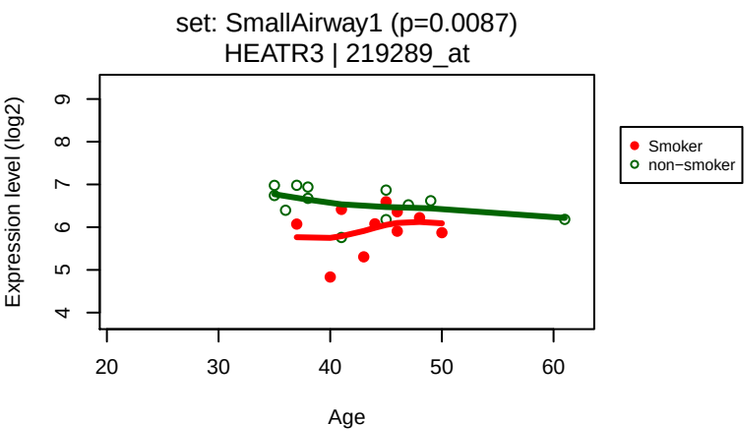
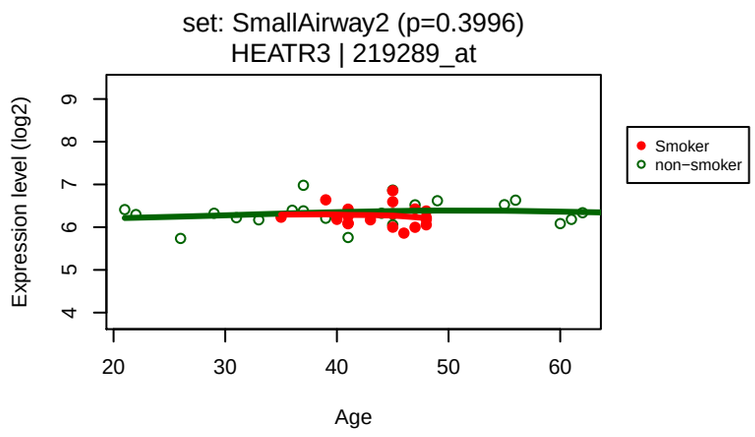
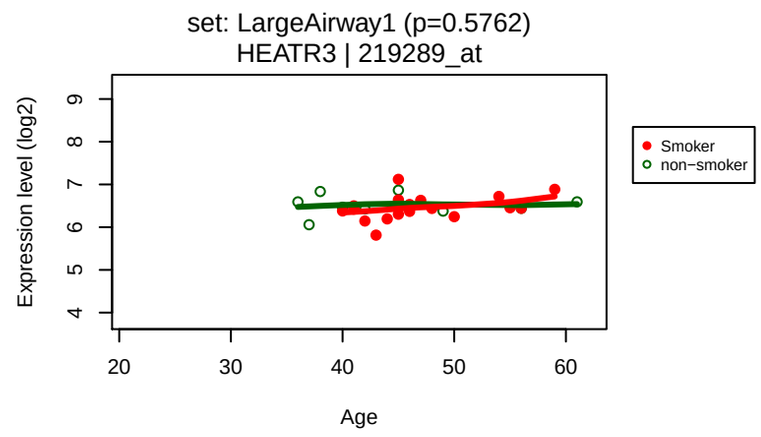

Gene expression of smokers and nonsmokers
gene: HEATR3 | 219289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0087
p=0.3996
p=0.1766
p=0.5762
p=0.413
p=0.1335
p=0.2371
p=0.3249
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

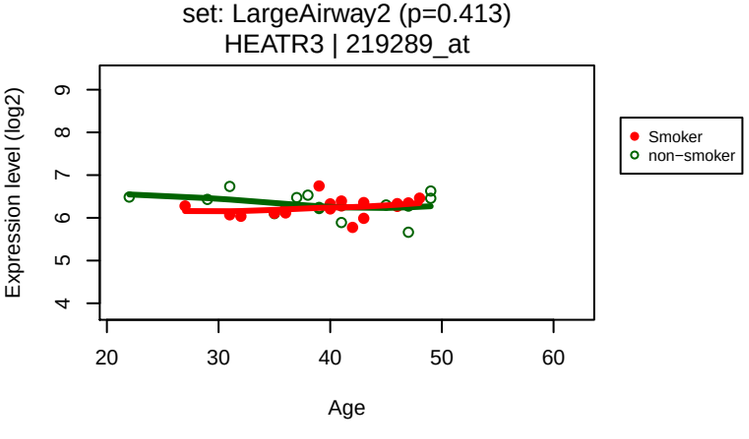

Gene expression ratio (smokers/nonsmokers)
gene: HEATR3 | 219289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0087
p=0.3996
p=0.1766
p=0.5762
p=0.413
p=0.1335
p=0.2371
p=0.3249
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

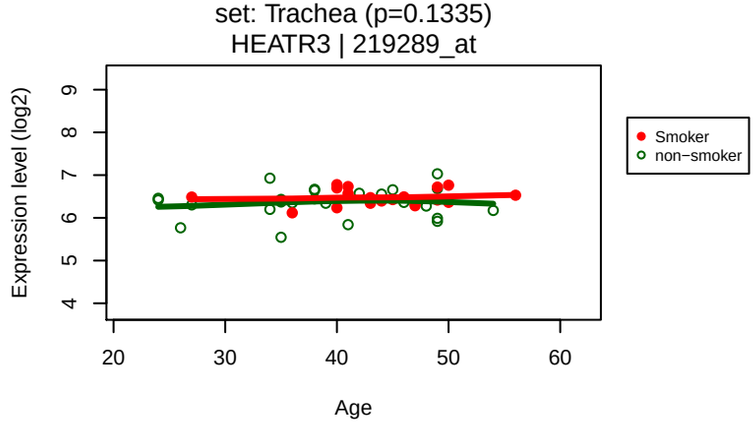

Gene expression ratio in Males
gene: HEATR3 | 219289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8633
p=0.5842
p=0.2921
p=0.9437
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HEATR3 | 219289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0243
p=0.4054
p=0.556
p=0.0596
SmallAirway2
LargeAirway2
Trachea
Macrophage2

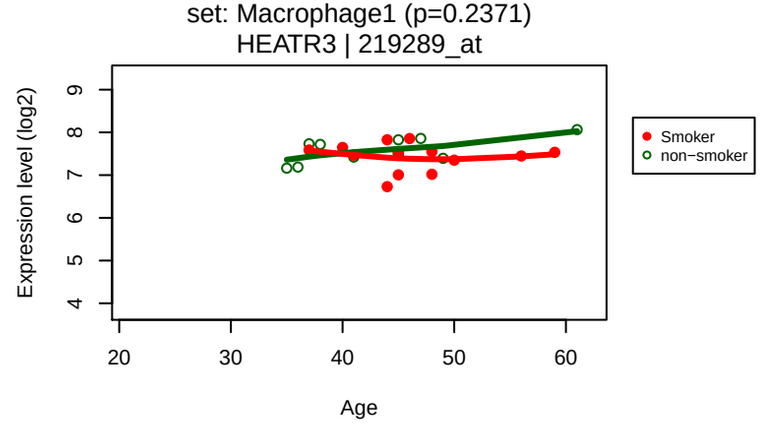

set: SmallAirway1 (p=0.0087)
HEATR3 | 219289_at
Expression level (log2)
Age
Smoker
non−smoker

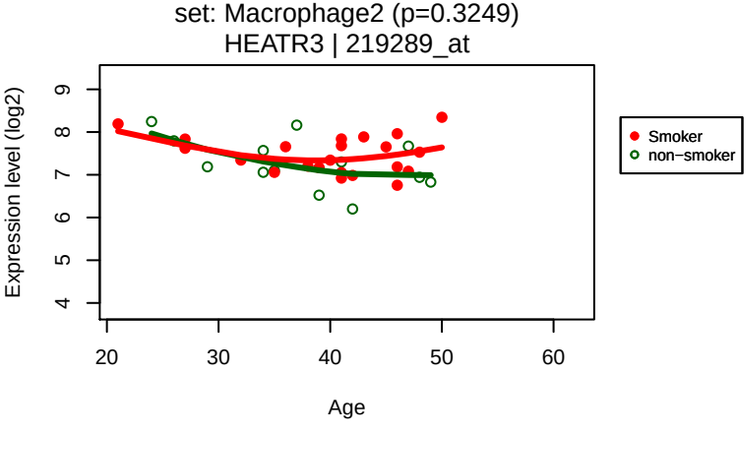

set: SmallAirway2 (p=0.3996)
HEATR3 | 219289_at
Expression level (log2)
Age
Smoker
non−smoker

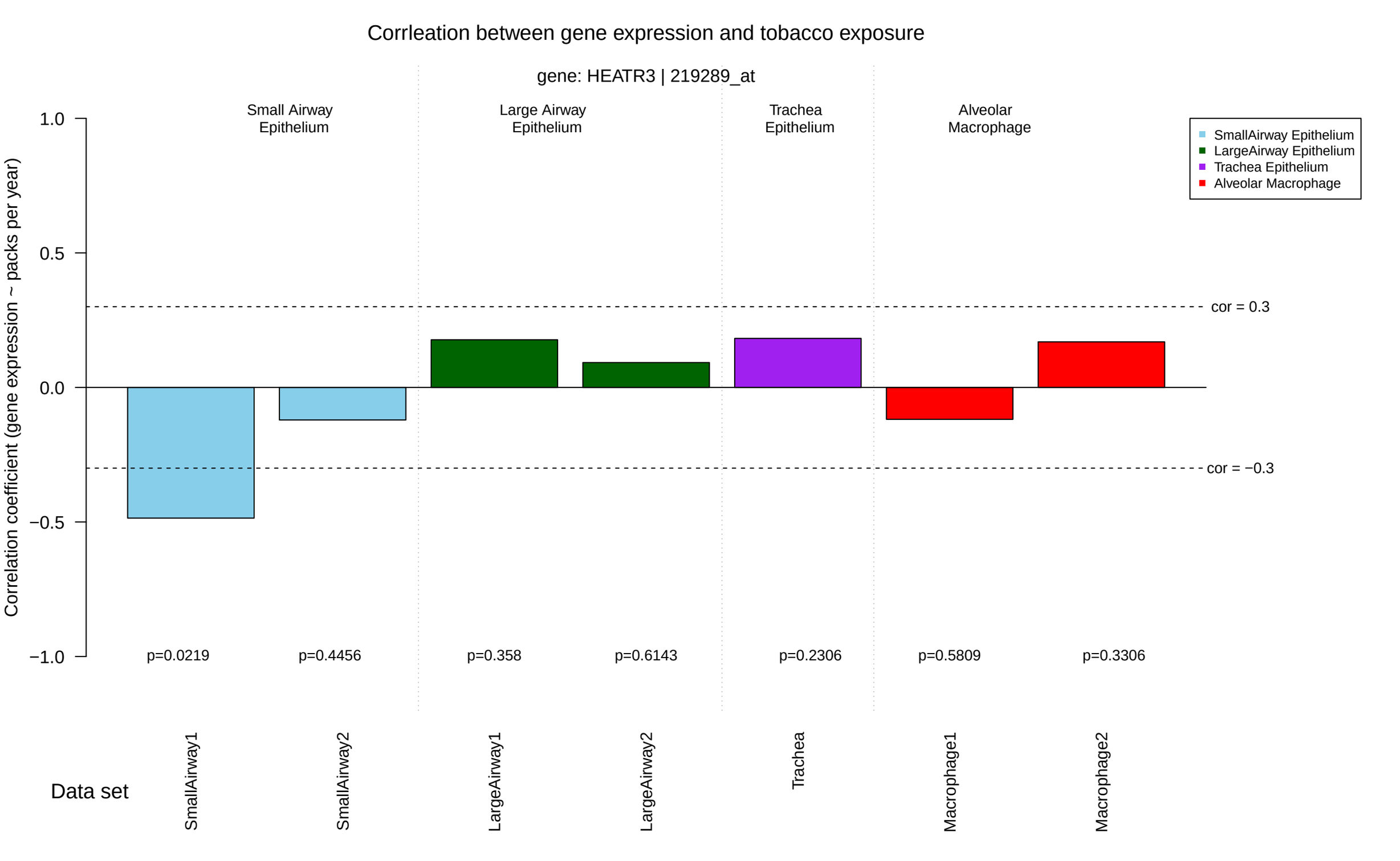

set: LargeAirway1 (p=0.5762)
HEATR3 | 219289_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.413)
HEATR3 | 219289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1335)
HEATR3 | 219289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2371)
HEATR3 | 219289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3249)
HEATR3 | 219289_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HEATR3 | 219289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0219
p=0.4456
p=0.358
p=0.6143
p=0.2306
p=0.5809
p=0.3306
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2